

Module 04 — External Cohort Validation

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{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE)
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Module 04 — External Cohort Validation

Part of the CausoBiome Pipeline

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Version: 1.0

Objective

This module validates microbial biomarkers in an independent metagenomic dataset (PRJEB10878). It evaluates: - Consistency of CRC vs. control trends across datasets - Statistical significance using Mann–Whitney U tests - Shared directionality and effect size - Visual agreement of biomarkers using plots and Venn diagrams

Script Overview

External_cohort_Validation.ipynb

Purpose:

Evaluate differential abundance and consistency of microbial biomarkers between internal and external datasets.

Inputs: - Filtered_biomarker_annotated.csv

- Internal: combined_ARG_VFDB_final_DATA.csv, Metadata_Aligned_VF_ARG_CountMatrix.csv

- External: Matrix_PRJEB10878_combined_VF_ARG.csv, PRJEB10878_Metadata_Aligned.csv

Key Steps: 1. Recode sample groupings (Case/Control, CRC/Control) 2. Perform Mann–Whitney U tests internally and externally 3. Compute mean differences (CRC - Control) per biomarker 4. Assess trend consistency and plot barplot 5. Venn and barplot of shared significance 6. Annotated scatterplot for consistent biomarkers 7. Save merged results with trends and flags

Output Files

Output File	Description
External_MannWhitney_Stats.csv	Stats from external CRC vs. control test
Internal_MannWhitney_Stats.csv	Stats from internal case vs. control test
Trend_Comparison.csv	Mean trend comparison table
Barplot_Trend_Comparison.png	Barplot of trends for shared biomarkers
Merged_Significance_Comparison.csv	Unified significance table with trend concordance

Output File	Description
Significant_in_Both_Datasets.csv	Biomarkers significant in both datasets
Significance_Comparison_Barplot.png	Barplot of overlapping significance labels
Significance_Comparison_Venn.png	Venn diagram of shared significance
Consistent_Trends.png	Annotated scatterplot of consistent biomarker trends

Visual Highlights

- **Barplot of Trends:** Displays direction of case/control mean differences for both datasets
 - **Venn Diagram:** Illustrates overlap of statistically significant biomarkers
 - **Annotated Scatterplot:** Shows correlated trends in internal and external datasets with size-encoded magnitude
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End of Module 04 README